

Quick Check

Lesson 6: Absorption, Respiration and Regulation: Explaining the Fall Back to 4.5 mmol/L

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Chemicals of Life | Lesson 6

Wanjiru's blood glucose reached a peak of 8.0 mmol/L at 60 minutes after her meal of ugali, sukuma wiki and beans. Suppose nothing removed glucose from her blood after that time. What would her blood glucose concentration do?

- A** It would keep rising without limit because glucose is made in the blood
- B** It would stay high, close to 8.0 mmol/L, because nothing is taking glucose out
- C** It would fall back to 4.5 mmol/L anyway because glucose slowly disappears
- D** It would fall because glucose is lost in sweat and urine

Wanjiru's blood glucose was 8.0 mmol/L at 60 minutes and 6.5 mmol/L at 90 minutes. By how much did it fall in this 30-minute interval?

A 3.5 mmol/L

B 1.5 mmol/L

C 0.5 mmol/L

D 14.5 mmol/L

Wanjiru's blood glucose rose by 3.5 mmol/L in the first 60 minutes (4.5 to 8.0). It then fell by 3.5 mmol/L from 60 to 180 minutes (8.0 to 4.5). How many times longer did the fall take than the rise?

- A 3.5 times as long
- B 2 times as long
- C The same time, because both changes are 3.5 mmol/L
- D Half as long, because the body removes glucose quickly

In the role-play of glucose after a meal, a learner holds up insulin cards when the blood glucose is high. Which statement correctly describes insulin?

- A** The pancreas releases insulin, and it causes cells, the liver and muscles to take up glucose from the blood
- B** The small intestine releases insulin, and it moves glucose into the blood
- C** Insulin is a food that gives the body energy when we eat
- D** The liver releases insulin, and it digests starch into glucose

Inside body cells, glucose is used in respiration to release energy. Which equation correctly shows respiration?

- A** Glucose + water gives starch + oxygen + energy
- B** Glucose + carbon dioxide gives oxygen + water + energy
- C** Glucose + oxygen gives carbon dioxide + water + energy
- D** Glucose + oxygen gives glycogen + carbon dioxide + energy

After a meal, some glucose is not needed straight away. What happens to this excess glucose?

- A** It is removed from the body as sugar in urine
- B** It is changed to insulin in the pancreas
- C** It is destroyed in the blood
- D** It is stored as glycogen in the liver and muscles

Learners write a **Claim-Evidence-Reasoning** explanation of Wanjiru's graph. Which piece of evidence best shows that the glucose in the blood came from digesting the starch in ugali?

- A** Her blood glucose fell by 3.5 mmol/L over 120 minutes
- B** The normal fasting range of blood glucose is 4.0 to 5.5 mmol/L
- C** Saliva containing amylase, kept at 37 degrees C with starch, gave a brick-red colour with Benedict's test after 10 minutes
- D** Sukuma wiki is a green vegetable

Which flow diagram shows the correct order of the final model that explains Wanjiru's blood glucose graph?

- A** Food chemicals, then digestion by enzymes, then glucose in blood, then cells (respiration) and storage, then back to normal
- B** Food chemicals, then glucose in blood, then digestion by enzymes, then cells and storage, then back to normal
- C** Digestion by enzymes, then food chemicals, then glucose in blood, then back to normal, then cells and storage
- D** Food chemicals, then cells and storage, then digestion by enzymes, then glucose in blood, then back to normal

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).